SMART INDIA HACKATHON 2025

VILLAGE GENTLE

- **Problem Statement ID –** SIH25010
- **Problem Statement Title-** Smart Crop

Advisory System for Small and Marginal Farmers

- **Theme-** Agriculture, FoodTech & Rural Development
- **PS Category-** Software
- **Team ID-** 55038
- **Team Name (Registered on portal) -** LastHope

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❖ Proposed Solution

Village Gentle is a comprehensive AI-powered agricultural advisory platform designed to bridge the digital divide for small and marginal farmers in India. The platform combines cutting-edge technology with multilingual support to provide personalized, real-time farming guidance.

- Our **Smart Farming Assistant** page empowers small and marginal farmers to ask questions in **their own language**—via **text or voice**—and receive real-time, AI-driven agricultural advice.
- Our **Crop & Fertilizer Recommendation** page provides farmers with personalized, data-driven guidance on the most **suitable crops** and **fertilizers** based on **soil nutrients**, **pH**, temperature, rainfall, and humidity.
- Our **Pest & Disease Detection** page enables farmers to **upload crop images** and instantly **detect** possible **infections** providing practical **treatment** steps. It also generates preventive care tips, educational videos, and relevant articles in the farmer's language.
- Our **Weather Advisory** page automatically captures the farmer's location using **GPS** and integrates real-time meteorological data with AI-powered insights to support smarter farming decisions.
- Also our **Economic Opportunity** page provides farmers with real-time **market price** updates along with guidance on business opportunities and skill development. This helps them make better financial decisions, access **new income sources**, and plan their sales more effectively.

Prototype Preview

❖ Innovation and Uniqueness:

- ✓ Web + Mobile app ensures accessibility across devices
- ✓ 7-in-1 Comprehensive Platform(Includes Rural Healthcare and Community Support)
- ✓ Ready to serve 600+ million farmers from day one
- ✓ Timely alerts for weather and market prices
- ✓ Multilingual support for regional accessibility

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TECHNICAL APPROACH

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- **Multilingual AI Chatbot & Voice Support:**
Mistral AI for context-aware responses, **AssemblyAI** for speech-to-text, **Google Translator API** for regional language translation, and **gTTS** for voice output in preferred language.

- **Real-time Location-Specific Advisory:**
Integrated **Open-Meteo API** for detailed weather forecasts, **Windy Maps** for live weather visualization, and **Open AI** for intelligent crop and farming advisories.

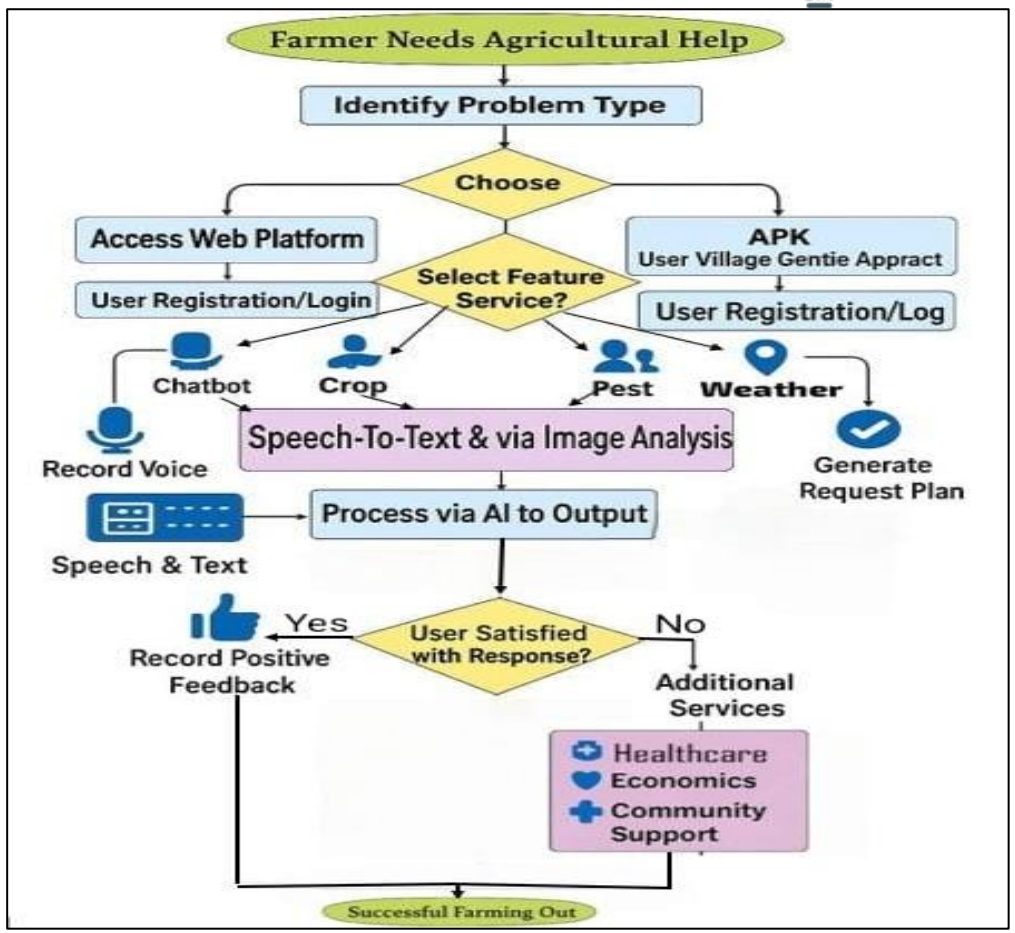
- **Soil health and Fertilizer guidance:**
Scikit-learn ML models for crop prediction, **decision tree models** for fertilizer recommendations, and **soil parameter analysis**.

- **Pest/Disease Detection:**
Kindwise Plant API for professional crop disease identification, **OpenCV** for image preprocessing, and **PyTorch** models for detecting crop-specific diseases.

- **Market Price Tracking:**
Google Custom Search API and **Open Market Data APIs** to provide real-time crop price updates across regions. **Geospatial APIs** map region-specific price variations.

- **Rural Healthcare and Community Support:**
Google Custom Search and **OpenRouteService** (ORS) with **Folium Maps** discovers nearby hospitals or clinics, **MongoDB Atlas** stores community issues, **Leaflet.js with Mapbox API** supports real-time geotagged reporting, **Twilio** handles SMS/email confirmations.

- **Other Technologies**
Flutter with WebView for cross-platform compatibility
Hosted on **Render.com** with auto-scaling.



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FEASIBILITY AND VIABILITY

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Feasibility:

1. Technology: Utilizes existing technologies (web, mobile, AI, API, Cloud)
2. Market: 86% of Indian farmers are small or marginal
3. Economic: Low operational costs due to cloud-based architecture
4. Operational: Continuous content updates through automated systems

Viability:

1. First AI-based platform providing personalized, real-time crop guidance
2. 68% of farmers use smartphones (FICCI-EY Report 2023)
3. ICT-based advisories can increase crop yield by 20–30%
4. Scalable advisory for Agri-tech startups

Challenges:

1. Small farmers' limited budget for technology services
2. Farmers' reluctance to trust new platforms without seeing results.
3. Low digital literacy and skepticism of technology
4. Difficulty in maintaining long-term user engagement.

Overcome:

1. Optimized for low-end smartphones and slow networks
2. Partnerships with local cooperatives, NGOs.
3. Village workshops and demo sessions, building trust step by step.
4. Regular personalized tips to keep farmers motivated

Business Potential:

1. 67% rural Indians use smartphones (2024)
2. Climate risks cut yields by 15–25%, need for adaptive farming
3. Government partnership potential
4. Agri-tech market will grow by $878M(2024) to $6.1B (2033)

Revenue Model:

1. Premium Subscription for advanced features
2. Advanced AI insights, priority support, detailed reports
3. Collaborations with Govt. agriculture, rural development departments
4. Anonymized data insights to policy makers, researchers, and agribusinesses.

Supporting Facts for Feasibility and Viability

- The **Indian Agritech market** is poised for explosive growth, from approximately **USD 878 million in 2024**, projected to reach **USD 6.15 billion by 2033**. Sourece: IMARC Group
- ICT-based advisory services resulted in incremental yield increases of approximately **25% in Bengal gram and sorghum**, and **12% in cotton**. Source: journaloffarmsciences
- Smartphone ownership among rural Indian households has more than doubled—from **36% in 2018 to 74.8% in 2022**. Source: Research Gate Publication

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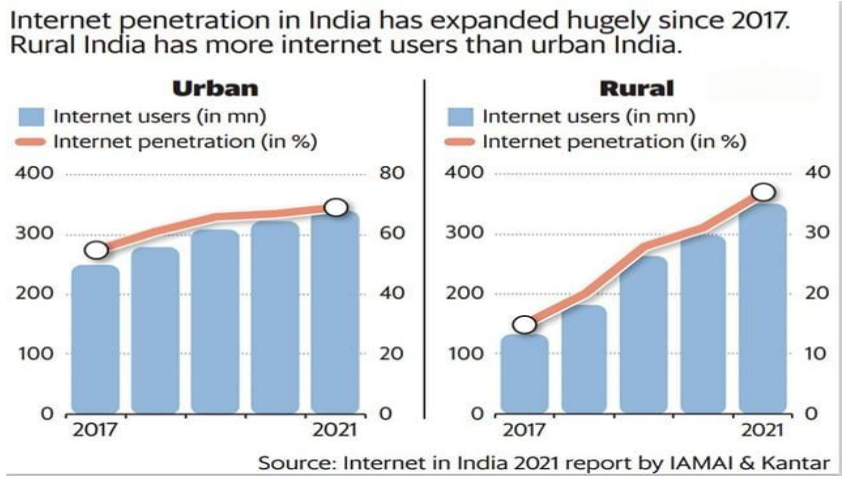
IMPACT AND BENEFITS



❖ Initial Impact on the target audience:

•**Freedom from Dependency:** Farmers gain independence from shopkeeper-driven advice.
•**Smart Cost Efficiency:** 15–25% fertilizer savings through precision recommendations.
•**Native-Language Simplicity:** Seamless communication in farmers' own languages.
•**Loss Control:** Early pest detection prevents 20–30% crop losses.
•**Empowered Decision-Making:** Data-driven insights replace blind guesswork.
•**Eco-Smart Practices:** Promotes sustainable and environmentally safe farming.
•**Seamless Governance Link:** Direct, digital access to agricultural schemes and authorities.

❖ Benefits:

Social Benefits:
- Attracts young people to agriculture through technology
- Strengthens rural communities through collective solutions
- Reduces migration to cities by making farming more profitable
- Empowers women farmers with equal access to advisory services

Economic Benefits:
- Better price realization through market information
- Better crop planning improves market supply
- Quality improvement enables international market access

Environmental Benefits:
- Reduces chemical exposure through optimized usage
- Better crop planning reduces post-harvest losses
- Smart irrigation advice reduces water wastage

Educational Benefits:
- Transfers research-based practices to traditional farmers
- Educates about credit, insurance, and investment options
- Provides ground-level data for policy formulation

Long-term Benefits:
- Better farming practices improve food quality and safety
- Agriculture-based industries in rural areas
- Major agricultural exporter globally

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RESEARCH AND REFERENCES

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❖ References

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2. "Digital Agriculture: Farmers in the Information Age"- FAO Report (2022)
3. "AI in Agriculture": A Comprehensive Review (2022)
4. "Voice User Interfaces for Rural Applications"- CHI Conference on Human Factors in Computing Systems
5. "Computer Vision for Crop Disease Detection: A Survey" - IEEE Transactions on Agricultural Engineering

Demo: Click Here

❖ Comparison with Existing Systems

Feature	Village Gentle	Kisan Suvidha	AgriApp	Plantix	Crop In
Multilingual Voice Support	✓ Native AI	✗ Limited	✗ Text Only	✗ Text Only	✗ English Only
Disease Detection	✓ AI-Powered	✗ Basic Info	✓ Limited	✓ Good	✗ No
Crop Recommendations	✓ ML-Based	✓ Basic	✓ Basic	✗ Limited	✓ Good
Weather Integration	✓ Hyper-local	✓ Basic	✓ Basic	✗ No	✓ Basic
Market Prices	✓ Real-time	✓ Basic	✗ No	✗ No	✗ No
Healthcare Support	✓ Compre-hensive	✗ No	✗ No	✗ No	✗ No
Community Platform	✓ Full Featured	✗ Limited	✗ No	✓ Basic	✗ No
Government Integration	✓ Ready	✗ Limited	✗ No	✗ No	✗ No
Mobile App	✓ Native + Web	✓ Basic	✓ Basic	✓ Good	✓ Basic
Personalization	✓ AI-Driven	✗ Limited	✗ Basic	✗ Limited	✓ Good
Continuous Learning	✓ Advanced	✗ No	✗ No	✗ Limited	✗ No